BANGGUO LI

Tsinghua University, Beijing, China
+86 15140375160 lbg21@mails.tsinghua.edu.cn

Education

Tsinghua University **Sep. 2021 – July 2025**
Bachelor of Engineering in Computer Science and Technology *Beijing, China*

GPA: 3.84/4.0(52/177), Sophomore Year: 3.90/4.0, Junior Year: 3.96/4.0

Coursework: Artificial Neural Networks, Fundamentals of Computer Graphics, Virtual Reality Technology, Introduction to Artificial Intelligence, Introduction to Machine Learning, Computer Organization, Software Engineering, Data Structures, Probability and Statistics, Introduction to Complex Analysis

Research

THU3DVICI Lab **June 2023 – Now**
Research Assistant *Tsinghua University, China*

Supervised by Prof. Li Yi, Institute for Interdisciplinary Information Sciences, Tsinghua University

- Co-conducted research on 3D comprehension and 3D generation (3D AIGC) via tuning foundational diffusion model to scalable 3D scene generaion.
- Tuning foundational image diffusion model with small-scale 3D dataset while utilizing multi-view images to achieve 3D generation. Utilizing geometric condition guided by internal 3D representation to improve 3D-aware consistency while preserving the creativity of foundational model.
- One paper submitted to CVPR 2024(Co-first author), focuing on scalable and generalizable 3D generation with small scale dataset tuning.

Experience

Student Association of Computer Science and Technology **2022 Fall – Present**
Vice Chair *Tsinghua University, China*

- One of the project leaders for the 27th THU Agent Competition titled AntWar, responsible for writing the core code.
- Led chapter of 15+ members to work towards the 28th THU Agent Competition, with hundreds of participants to date.

SAST 2023 summering Traning **2023 Summer**
Lecturer *Tsinghua University, China*

- Lectured Unity-related content and courses, with a total viewership of over several thousand.

Projects

VR-Gaussian | *Unity* **2024 Fall**

- Rendering gaussain scene representation in Oculus Quest 2, supporting rendering of Gaussian scenes, enabling roaming and scene switching within the environment.
- Rewrote shaders in Unity according to the original paper to support rendering Gaussian scenes on VR platforms.

RISC-V CPU | *SystemVerilog* **2024 Fall**

- Implemented a RISC-V compatible seven-stage pipeline CPU using SystemVerilog.
- Featuring support for interrupts, caching, and instruction prediction and successfully running UCore.

Crowdsourcing Platform | *JavaScript, Python* **2023 Fall**

- Implemented a crowdsourcing platform website based on React and Django, supporting features such as order posting, task completion, order review, and user recharge.

Award

Academic Excellence Award **2023 Spring**
Department of Computer Science and Technology, Tsinghua University
18 out of 177 were awarded

Outstanding Student Association Core Member **2023 Spring**
Tsinghua University

Technical Skills

Languages: Python, C++, C, SystemVerilog
English: IELTS 7.5, capable of academic writing and fluent in communicating in English
Skills: Pytorch, Machine Learning Frameworks, Computer Vision, Computer Graphics